RAKSHA-X
Autonomous AI-Powered Women Safety Bracelet
Zero-Network Protection Anywhere



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Why current solution fails?

Addressing the critical gaps in current security measures where network connectivity and manual phone access are often unavailable.



1. Women's safety is a major **global issue.**
2. Assaults often occur in **Low network zones**, Nighttime, and Isolated areas.
3. Most safety apps require a mobile phone, internet connection, and **manual activation.**
4. During panic, the victim may not access their phone or press an emergency button.
5. We need a fully autonomous, wearable, **mobile-free** safety solution.

RAKSHA-X – Proactive Safety Bracelet



1. RAKSHA is a **Standalone Smart Safety Bracelet** and AI-powered Threat Detection Device.
2. Works without a mobile phone and **without the internet**.
3. Biometrically locked to the user for maximum security.
4. Auto-alerts police and trusted contacts immediately.
5. Activates nearby Raksha users using a **Swarm Model**.

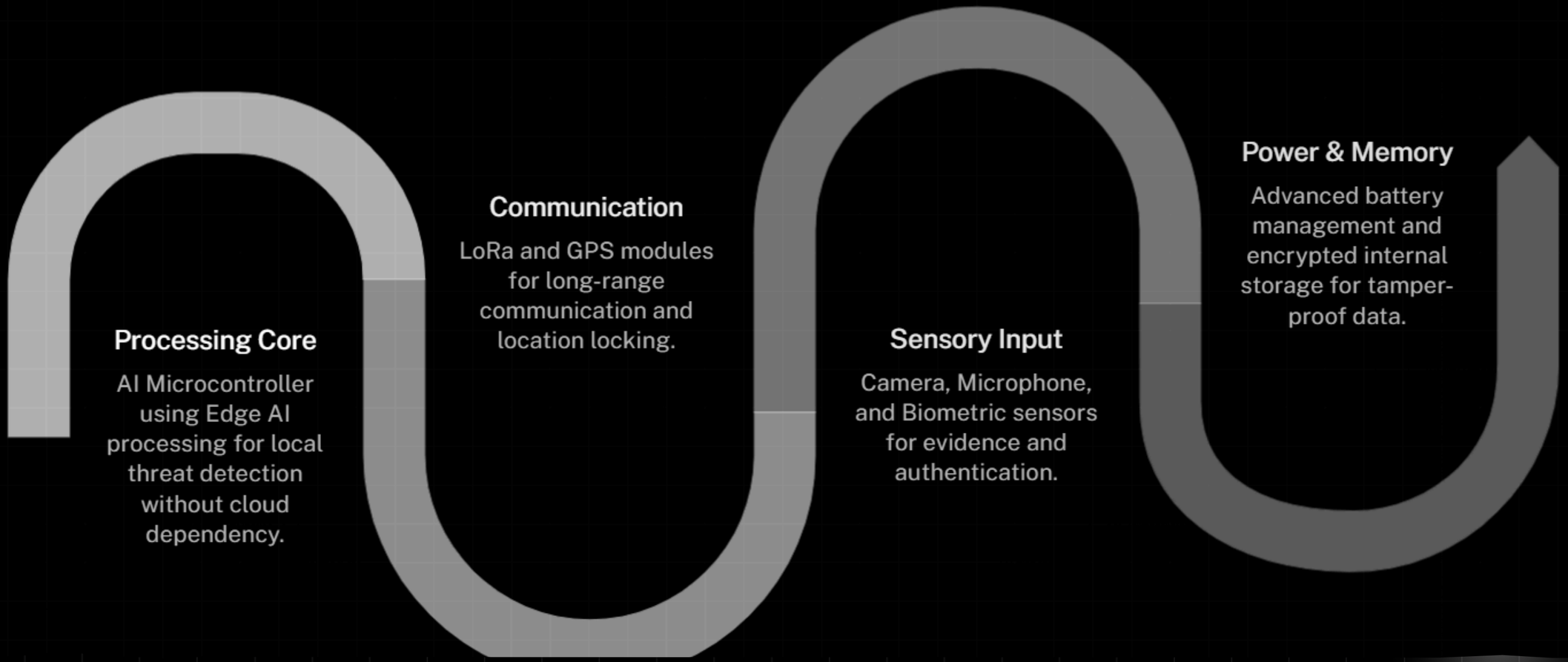
Key features

A comprehensive suite of autonomous safety tools integrated into a tamper-proof wearable device.

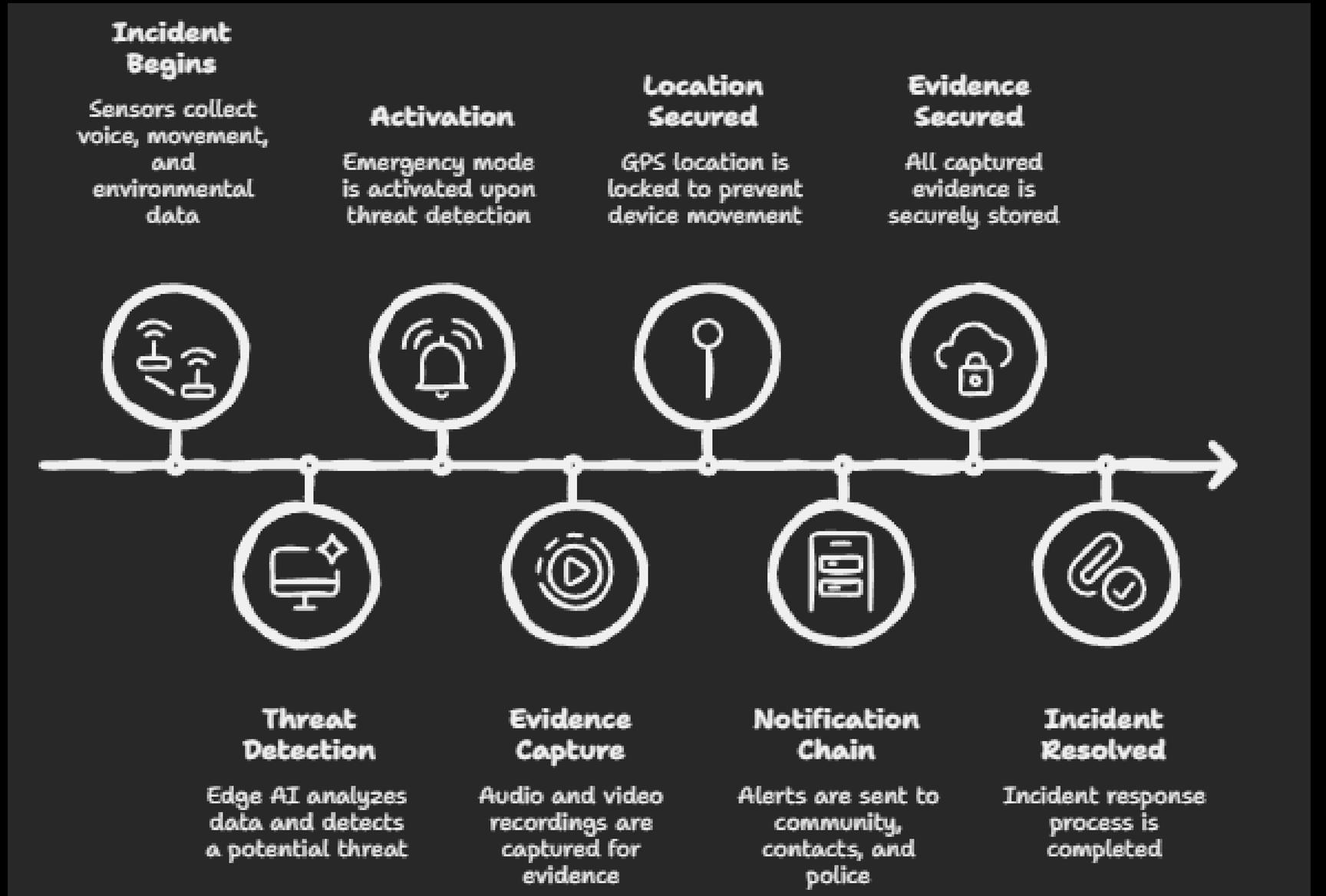
- AI-based Threat Detection
- Voice + Video Recording
- Offline Emergency Alert System
- Direct Police Alert & Trusted Contacts Auto Call
- Swarm-based Community Alert
- Biometric Lock & Tamper Detection
- Offline Mode via **Mesh Communication**

RAKSHA-X Architecture

A robust hardware ecosystem designed for low power consumption and high reliability without cloud dependency.

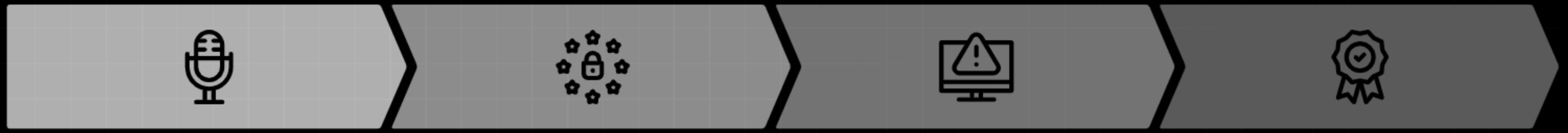


FLOW CHART DAIGRAM



Emergency mode-what happens?

A multi-stage autonomous response starts immediately upon threat detection to secure the user and gather evidence.



Record Evidence

Continuous Voice and Hidden Video Recording starts immediately to capture the situation.

Lock Location

GPS Location is locked to provide real-time tracking for rescuers and police.

Broadcast Alert

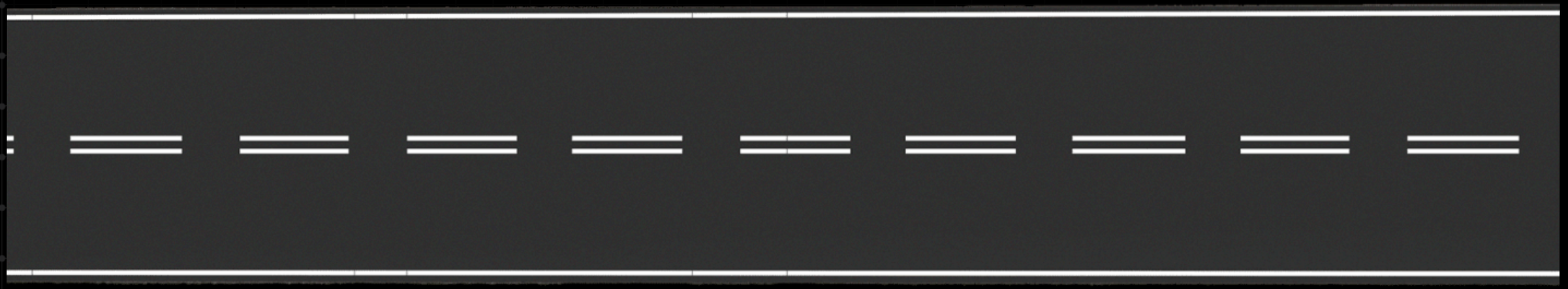
Real-time alert broadcast begins, including an optional loud alarm to deter threats.

Safety Validation

Recording stops only when the user reaches a safe zone or validates safety via biometric unlock.

How Threat Detection Works

RAKSHA uses Edge AI to continuously monitor patterns and activate emergency response without manual intervention.



1

Voice Patterns

- Monitors Screaming, Crying, and Panic tones
- Analyzes patterns locally via **Edge AI**

2

Movement Detection

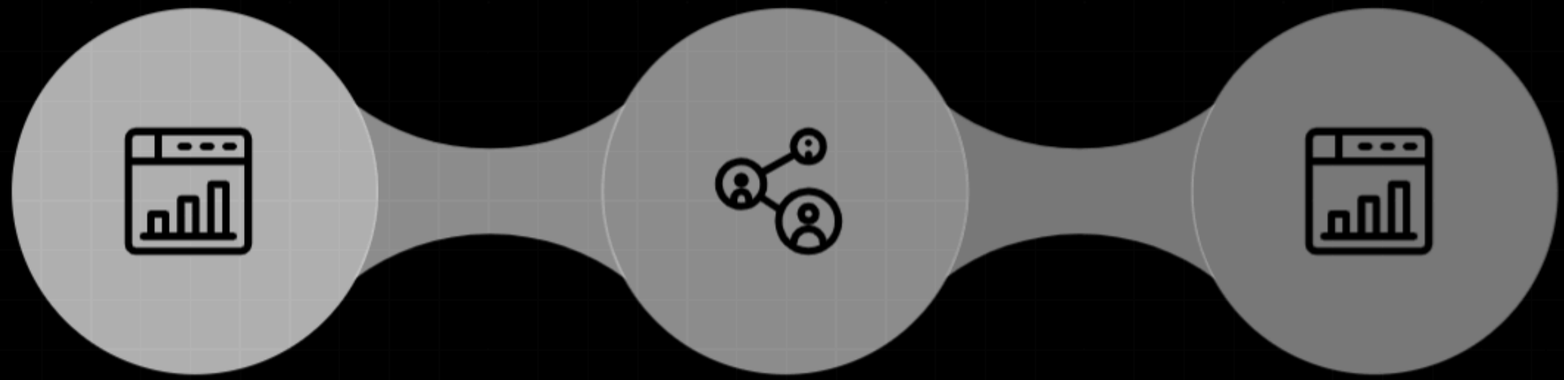
- Detects falls, forceful shaking, and running
- Triggers alert if threat probability > threshold

3

Environmental Change

- Darkness detection and unknown voices
- **No manual button** required for activation

Multi-level alert system



Level 1: Community

Immediate alerts sent to all nearby Raksha users for the fastest possible local deterrence.

Level 2: Contacts

Trusted contacts receive automatic calls and live GPS location details.

Level 3: Authorities

Nearest Police Station receives live location, audio evidence, and device ID for fast response.

Secure evidence storage

RAKSHA automatically encrypts and protects all recorded evidence for future legal use, preventing manual deletion.

Audio and video data are **fully encrypted** during emergencies

Recordings **cannot be deleted manually** to prevent tampering



Stored in secure internal memory with auto-upload capabilities

Provides tamper-proof digital proof for legal proceedings

Future Scope

- Government integration with national emergency databases.
- University campus and school safety network versions.
- Smart-city integration for a cross-city safety network.



Thank you